



Image: <https://www.mlb.com/video/josh-hader-k-s-the-side-during-his-astros-debut>

How Should We Evaluate Relief Pitchers in MLB?

In the modern MLB, relief pitchers have become increasingly important parts of a team's pitching staff. Starting pitchers are throwing fewer innings as a whole, and the burden of these extra hundreds of innings falls on the bullpen. At the same time, this position is one of the most volatile, and any given reliever can experience major deviations in their effectiveness between seasons. For example, closers such as Josh Hader and David Bednar, players who have been successful in the past few years, have struggled at the outset of the 2024 season. On the other hand, Emmanuel Clase has returned to an elite level following a subpar 2023 season that now appears to be an outlier season. These big-name closers garner the most attention due to their role, and most people would argue that a reliable closer is one of the most valuable assets on a team, but the rest of the bullpen should not be ignored either.

With how unpredictable the closer position has been, some teams have opted to shift toward a "closer by committee" approach, deploying relievers by matchup and situation rather than in fixed roles. This also challenges the conventional wisdom surrounding the value of the Save statistic, which has traditionally been used to value relief pitchers. However, many teams have continued with the traditional approach successfully, indicating that strategy is still very roster-dependent. Nonetheless, this illustrates the

importance of considering the value of all relief pitchers, not just closers, by considering statistics that apply more broadly than Saves.

The question of which statistics are best fit to analyze relief pitchers depends greatly on the context. Generally, WAA (Wins Above Average) and WAR (Wins Above Replacement) are used to quantify the value a player contributes. However, these are used less with relievers since their emphasis on volume undervalues the lower number of innings pitched by relievers. Alternatively, WPA (Win Probability Added) measures the impact of a reliever by summing their contributions to win probability in each appearance. This approach tends to value innings in higher-leverage situations, not necessarily weighing the sheer number of innings as heavily. It is also useful to consider how these advanced metrics translate with basic stats such as ERA (Earned Run Average) and WHIP (Walks + Hits Per Inning Pitched), as these can be calculated without a complex approach.

My final goal with this project is to develop a model that better understands the volatility of relief pitching by studying career trends and game logs for individual pitchers. This way of quantifying and predicting success could be very applicable to analyzing potential trades and free-agent signings, while also offering a tool in fantasy baseball and season projections. This first step toward that goal involves determining the statistics that best correlate with success. Using WAA, WAR, WPA, RE24 (Run Expectancy Saved), and PchR (Pitching Runs) as baseline metrics for success, I computed correlations of 25 other relatively basic stats with these to identify the strongest indicators of success.

The dataset that was analyzed in this project included 138 relief pitchers from the 2023 MLB season. These pitchers all made over 40 relief appearances and pitched every game in relief. The 25 explanatory stats were primarily drawn from the pitching stats on Baseball Reference, with a few hybrids that were calculated using simple formulas from the available pitching stats. Using this set of 25 statistics that encompassed a wide range of aspects of pitcher effectiveness, from strikeouts to run prevention, a correlation matrix was computed using the Pandas library in Python between the metrics for success and explanatory stats.

Stats:

The majority of stats that were used were directly taken from Baseball Reference, and their definitions are included there. The following stats were calculated by combining data from related basic stats.

- GF% (Percent of Games Finished): $\text{GF (Games Finished)} \div \text{G (Games Played)}$

- SV+H (Saves + Holds), SV+H%: Sum of Saves and Holds, divided by SVSit (Save Situations) for the percentage
- LERA+ (Leverage ERA): Product of ERA+ and aLI (Average Leverage Index) divided by 100
- S/K% (Swinging Strikeout Percentage): Product of SO% (Strikeout Percentage) and S/SO% (Strikeout Swinging Percentage)
- TBIP (Total Bases per Inning Pitched): TB (Total Bases) + Walks, divided by the number of innings pitched

Results:

The correlations between the five baseline metrics were strong, but WPA clearly had a weaker correlation with the other four. With the lowest correlation still having $R > 0.7$ (between WPA and PtchR), these five metrics are very consistent in measuring success. The slight differences are due to the different calculation methods of each. WAA, WAR, and PtchR place less emphasis on the game situation, so their strong correlations are unsurprising. On the other hand, WPA and RE24 are inherently situational stats but focus on different elements (win probability vs run probability). Overall, the strong correlations between these metrics support the idea that slightly different definitions of success still agree. Future analysis could aggregate these into one metric for success, but this project generally treats them separately.

	WAA	WAR	WPA	RE24	PtchR
WAA	1.000000	0.983754	0.732152	0.880642	0.962330
WAR	0.983754	1.000000	0.772662	0.878276	0.947814
WPA	0.732152	0.772662	1.000000	0.791683	0.707415
RE24	0.880642	0.878276	0.791683	1.000000	0.878405
PtchR	0.962330	0.947814	0.707415	0.878405	1.000000

The correlations of each of the explanatory stats were computed with these five metrics, and an overall ranking was created by considering the average magnitude of R-values. These averages generally agree with the strengths of each individual metric, and most of the deviation was very slight. However, there were a few notable differences that will be discussed below.

RA9	0.868023
ERA	0.846043
ERA+	0.774543
TBIP	0.751022
LERA+	0.737333
OPS	0.729733
WHIP	0.692690
OBP	0.651118
SLG	0.648998
SO-BB-HR-H9	0.610247
H9	0.588400
BA	0.572810
SV+H	0.563732
FIP	0.526942
aLI	0.470244
SO-BB-HR%	0.440627
SO	0.403125
SV+H%	0.398102
S/K%	0.394148
SO%	0.391579
SO-BB%	0.384902
HR9	0.384066
SV	0.343057
GF%	0.231479
BB%	0.129304

From this ranking, the strongest explanatory stats are RA9, ERA, and ERA+, an unsurprising finding considering that runs are ultimately what determines the outcome of a baseball game. The success metrics are thus likely to assign significant weight to runs, as a failure to prevent runs renders all other elements of an outing less significant. Interestingly, though, adding leverage through LERA+ weakened its correlation with the metrics overall (although it remains strong). This is evidence for the idea that many of these metrics do not consider the game situation much, and the findings for WPA strongly support this. LERA+ was the only strong correlation with an R-value > 0.7 (R = 0.717), significantly greater than the other run stats. WPA also showed notable differences with other stats, but this result clearly showed the comparison between WPA and the other metrics.

	WAA	WAR	WPA	RE24	PtchR
RA9	0.938055	0.909140	0.667695	0.865826	0.959401
ERA	0.905849	0.884581	0.654004	0.828023	0.957755
ERA+	0.819238	0.810237	0.649042	0.747965	0.846232
LERA+	0.744251	0.788695	0.716875	0.698519	0.738325

Of the stats that do not directly consider runs, TBIP and opponent OPS were the only two strong correlations. As analogs that measure the same production from a pitcher and batter perspective, this is not shocking. While not directly related to runs, OPS is one of the best indicators for offensive production, so its strong correlation to pitcher success makes sense. WHIP and opposing OBP and SLG were also not far behind these two, as three stats that are more intuitive to contextualize by disaggregating TBIP and OPS. These findings are also not shocking, but they do provide further evidence in the debate between using AVG and OPS for studying batters.

	WAA	WAR	WPA	RE24	PtchR
TBIP	0.766314	0.752965	0.623231	0.812241	0.800362
OPS	0.749633	0.730135	0.602537	0.783274	0.783089
WHIP	0.703177	0.685155	0.560156	0.769839	0.745120
OBP	0.659916	0.640969	0.522805	0.724799	0.707100
SLG	0.671466	0.654937	0.543649	0.683403	0.691534

As mentioned above, the ranking under WPA gave the greatest deviation, clearly favoring the inclusion of leverage and game situation. While illustrated through LERA+, the most notable difference that showed this was the correlation of SV+H being the 6th highest with WPA, only behind the run stats and TBIP. Comparatively, SV+H was 13th on average, representing one of the largest differences. Other large differences include

aLI (10 vs 15), SV+H% (13 vs 18), and SV (17 vs 23). The emphasis on Saves and leverage is clear through these stats, but it is also worthwhile to note the mechanisms behind the shifts.

LERA+	0.716875
RA9	0.667695
ERA	0.654004
ERA+	0.649042
TBIP	0.623231
SV+H	0.603893
OPS	0.602537
WHIP	0.560156
SLG	0.543649
aLI	0.529339
OBP	0.522805
SO-BB-HR-H9	0.519357
SV+H%	0.503875
H9	0.484159
FIP	0.467436
BA	0.464844
SV	0.404252
SO-BB-HR%	0.380205
SO	0.376312
S/K%	0.372337
SO%	0.358484
HR9	0.351211
SO-BB%	0.328020
GF%	0.271933
BB%	0.092229

The correlations with WPA were significantly weaker across the board, thus allowing exceptions to be very noticeable. LERA+ exhibited a lesser deviation from average (-0.02) than other run stats (-0.2), allowing it to emerge as the strongest correlation. On the other hand, SV+H (+0.04), aLI (+0.06), SV+H% (+0.1), and SV (+0.06) all had stronger correlations with WPA than the other metrics, illustrating the weight that WPA uniquely gives to these situations. However, it is important to note that, despite these relative increases, the correlation strength between these stats and WPA was still not as strong as many other stats.

The weak correlations in many of the stats also showed interesting results. Most surprising was the weakest correlation for BB%, especially considering the conventional belief that walks are very dangerous for pitchers. The weak correlation for GF% is less surprising since it values all games equally, but the weak correlations with SV and SV+H% are more intriguing. Even with the increases when considering WPA, this seems to challenge the added value that many place on closers and late-inning relievers. However, this statistical analysis does not account for any human element and psychology of those situations, undervaluing pitchers who can perform under stress.

The moderately weak correlations with many strikeout and home run-related stats (SO-BB%, HR9, SO%, S/K%, SO-BB-HR%) are also surprising. One of the most coveted skills for relievers is the ability to miss bats, and pitchers who pitch to contact are becoming increasingly rare. However, this data contradicts that view, showing that a high strikeout rate is not as impactful as commonly believed. Additionally, the weak correlation with home run rate is unexpected due to the amount of impact one home run can have. Anecdotally, MLB teams are still likely to consider these “three true outcomes”, favoring pitchers with this skill set in individual situations due to the lesser variability compared to balls in play. Nonetheless, these results do not strongly illustrate benefits over a larger sample size across a full season.

A related result was the moderate correlation of FIP (Fielding Independent Pitching), generally regarded as one of the best predictive stats for pitchers. This statistic calculates the performance of a pitcher without considering defense by only incorporating strikeouts, walks, home runs, and hit-by-pitches. Given the weak correlations of these individually, it is unsurprising that FIP was also only moderately correlated, but it raises a question of consistency. While this does not necessarily mean that the metrics used in this analysis are flawed, and it also does not discredit the power of FIP, it does illustrate a source of incompleteness in the selected metrics. By only considering the outcomes a pitcher is responsible for, any variance in defensive play is eliminated, giving a more accurate predictor of success under a large sample size. The consideration of one season gave a reasonable sample size for this project, but the power of FIP will likely be more apparent upon extending this analysis to multiple years.

Conclusions:

Overall, this project emphasized the importance of run-based stats in determining success. The 5 metrics for success that were used provided a diverse yet consistent baseline that each of the explanatory stats was tested with, and the results largely matched expectations. A few of the surprising findings were the lack of correlation between traits that are generally valued highly, including strikeout ability and

performance in high-leverage situations. While the results showed that there was little correlation using these selected metrics, these are things that should be considered further.

After completing this first step, the next goal of this project is to use these results and develop a model for predicting and evaluating relief pitchers, building a stronger understanding of this important yet volatile position. Similar analyses can be extended to multiple seasons and the career spans of notable pitchers, and observing common trends could indicate predictors of success. Aggregating the valuation of success into one quantity will also be important to standardize the comparisons between pitchers. The differences that were observed between the selected metrics, especially WPA, indicate that this should be done carefully, and some consideration for real applications should also be included beyond a purely statistical analysis. The final model will be a useful tool for analyzing pitcher transactions and projecting overall team bullpen performance, with particular utility in the offseason and around the trade deadline as teams look to reconfigure their pitching staffs.